

DNS & Infrastructure Walkthrough: Personal Domain

Assignment: PF-04 | Track: General AI Fluency | Author: Muhammad Akbar Pradana

Live Host: akbarprdna.my.id / akbarpradana.netlify.app | Target Subdomain: akbar.flyrank.ai

1. Executive Summary & Core DNS Concept

The Domain Name System (DNS) functions as the phonebook of the internet, translating human-friendly names (e.g., akbar.flyrank.ai) into machine-readable IP addresses (e.g., 104.21.45.12). This document outlines how custom subdomains and CNAME records route web traffic securely over HTTPS.

2. What is a CNAME Record?

A **CNAME (Canonical Name) Record** maps one domain name (an alias) to another domain name (the canonical host). Unlike an A Record (which points directly to a fixed IP), a CNAME points to a destination domain, allowing the underlying hosting infrastructure to change IP addresses without breaking domain links.

Host / Subdomain : akbar.flyrank.ai
Record Type : CNAME
Points To / Value: akbarprdna.my.id (or akbarpradana.netlify.app)
TTL : 3600 seconds (1 hour)

3. The 4-Step DNS Resolution Lifecycle

When a user types https://akbar.flyrank.ai into their browser, the following sequence occurs within milliseconds:

Step	Component	Action & Resolution Detail
1	Local Browser & Resolver	Browser checks local cache; if missing, queries Recursive Resolver (ISP/Cloudflare 1.1.1.1).
2	Root & TLD Nameservers	Resolver queries Root server (.) -> referred to .ai TLD server -> referred to FlyRank DNS.
3	Authoritative Server	FlyRank authoritative DNS returns: akbar.flyrank.ai CNAME akbarprdna.my.id.
4	TLS Handshake (HTTPS)	Browser connects to host IP over port 443, validates SSL certificate (padlock ) , and loads page.

4. Capstone Subdomain Deployment Checklist

- **Live Verification:** Personal portfolio confirmed live and serving over HTTPS on clean domain.
- **Essential Links:** Profile positioning, LinkedIn, GitHub, CV download, and booking link verified.
- **Ops CNAME Provisioning:** FlyRank Ops provisions CNAME akbar -> host.domain upon capstone approval.
- **Host Binding:** Custom domain added in host settings with automated Let's Encrypt SSL padlock.